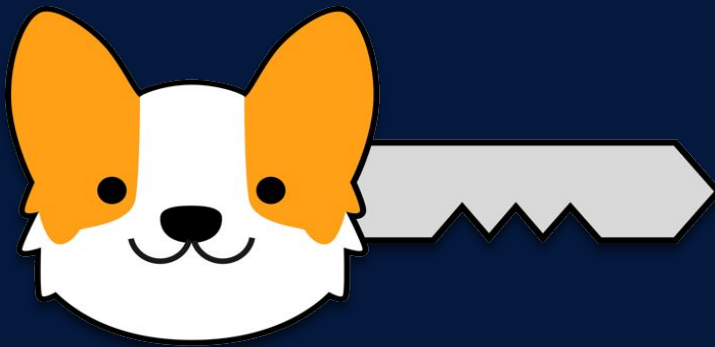


CorgiKey



Adam Yu, Arjun Singh, Hunter Massey, Olivia Gillam, Simon Yu

By Team Fat Stacks & Allthenticate



COMPUTER SCIENCE
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Allthenticate

MEET THE TEAM



Adam Yu (Lead)
Cross-Platform Dev



Simon Yu
Mobile Dev



Olivia Gillam
Mobile Dev



Arjun Singh
Authentication



Hunter Massey (Scribe)
Authentication

YOUR PASSWORD IS WEAK

Insecure

- Can be stolen by malicious actors through **phishing attacks** and **data breaches**
- Can be bypassed via SIM swapping & other types of attacks

Inconvenient

- Users burdened with **password memorization & management**
- Password restrictions (e.g. requiring digits, caps, symbols) make the process of coming up with a password cumbersome

Slow

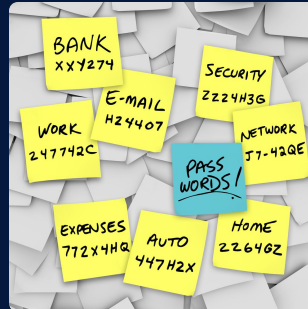
- Two-Factor Authentication (2FA) adds security at the cost of convenience
- Time-consuming process of resetting a password

The Uber Hack Exposes More Than Failed Data Security

Sept. 26, 2022

LastPass says it was breached — again

Zack Whittaker @zackwhittaker / 5:17 PM PST • November 30, 2022



UCSB

Choose an authentication method

Duo Push RECOMMENDED	Send Me a Push
Passcode	Enter a Passcode

☐ Remember me for 10 hours

Secured by Duo

ABOUT 2FA...

According to Uber, having obtained the contractor's password, the hacker sent repeated log-in requests to the contractor's account and was then able to bypass Uber's two-factor log-in authentication—a system where a user is granted access after electronically confirming their identity twice—when the contractor finally accepted the authentication. The hacker was also admitted to the Uber Slack account and posted a message that read: “I announce I am a hacker and Uber has suffered a data breach.”

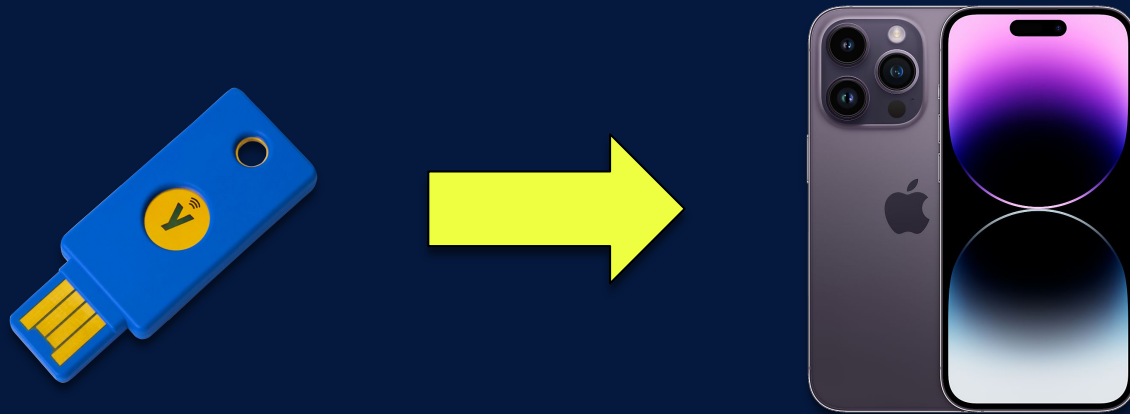
<https://www.bu.edu/articles/2022/what-you-need-to-know-about-uber-data-breach/>

OUR SOLUTION? REPLACE PASSWORDS

CorgKey is a mobile application that acts as a **mobile roaming FIDO authenticator**

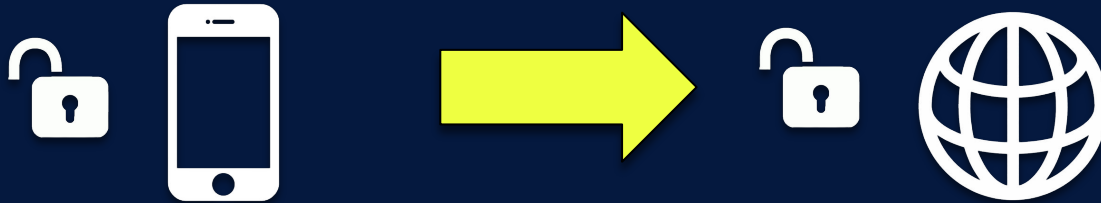
Imagine the **security** of hardware authenticators like Yubikey...

... wirelessly built right into your phone!

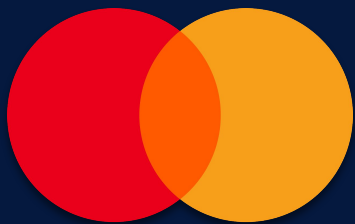


HOW DO WE DO IT?

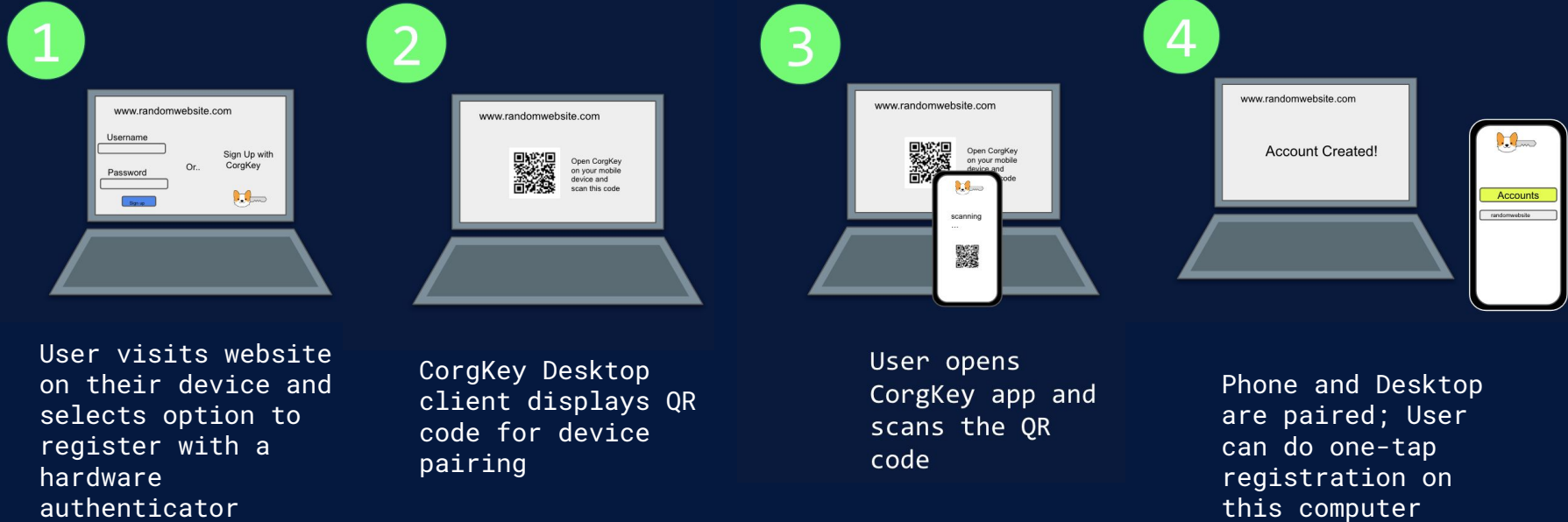
- **Implement the FIDO Protocol**
 - **What does that mean?**



FIDO BACKERS



User Flow (account creation)



User Flow (account login)

1



User visits website on their device and selects option to sign in with a hardware authenticator

2



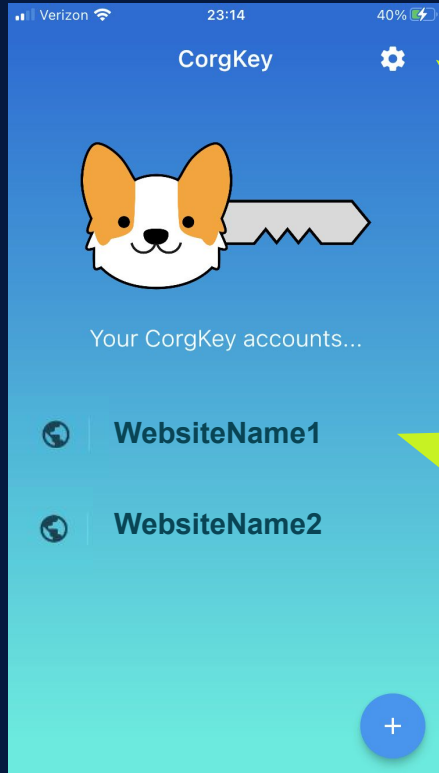
User accepts login attempt on CorgKey phone app

3



User is logged in

UI Overview



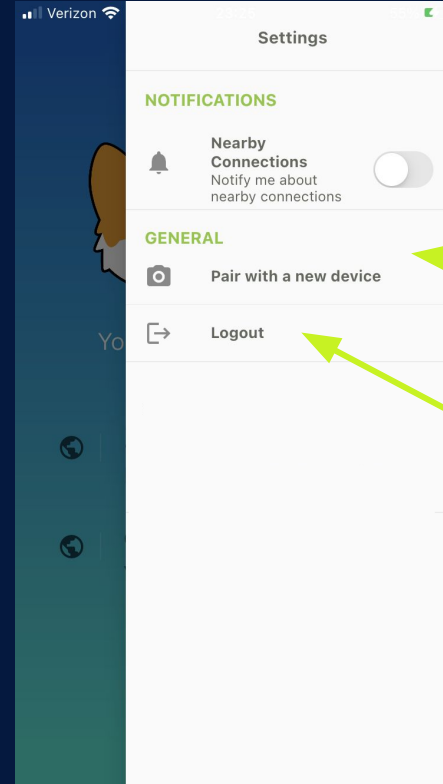
Open Settings
drawer



Name of website



WebsiteName1



Toggle
notifications



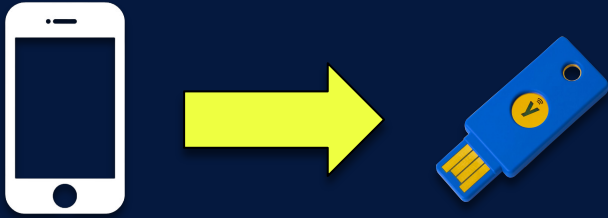
QR Code
Scanning
(for pairing)



Log out

CHALLENGES

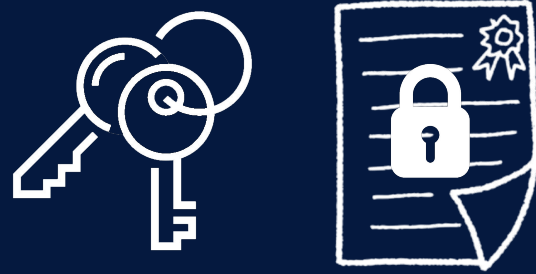
- Turning a phone into an Authenticator



- Pivoting from USB to FCM



- Certificate Encoding



- Inter Process Communication



UTILIZED TECHNOLOGIES



DEMO

**Security of Hardware
Authentication**

+

**Convenience of a
Mobile App**



Your password is weak

#DownWithDUO

The future is passwordless



QUESTIONS?

SPECIAL THANKS

- Bernie Conrad, Maria Matthew, Chad Spensky & Rita Mounir from Allthenticate!
- CS189A/B Course Staff
- Kelly Yan for the amazing logo!



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